

PAPER_972: ALICE Centrality Multiplicity via $S_{26}^{(k)}$

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216 **Source:** qgp_ramanujan_application.py (ALICECentralityMultiplicityCalculator) **Calculator:** ALICECentralityMultiplicityCalc (CP4 #556) **CVW:** v2.0.0 compliant

Abstract

We model the ALICE charged-particle pseudorapidity density $dN_{\text{ch}}/d\eta$ as a function of centrality and collision energy, modulated by the expanded Ramanujan factor $S_{26}^{(k)}$. This extends PAPER_364 with higher-order Ramanujan acceleration.

1. Multiplicity Formula

$$\frac{dN_{\text{ch}}}{d\eta} = A \cdot (\sqrt{s})^{0.156} \cdot \left(1 - \frac{c}{100}\right)^{\alpha} \cdot S_{26}^{(k)}([SSq])$$

where $A = 2.0$, $\alpha = 1.2$, c = centrality percentile (0% = most central).

2. Centrality Dependence

Centrality (%)	$dN_{\text{ch}}/d\eta$ (13.6 TeV)
0 (most central)	Maximum
5	$\sim 0.94 \times \text{max}$
20	$\sim 0.78 \times \text{max}$
50	$\sim 0.50 \times \text{max}$
80 (peripheral)	$\sim 0.19 \times \text{max}$

3. Energy Scaling

The $\sqrt{s}^{0.156}$ power law is consistent with ALICE Run 3 Pb-Pb data.

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
 2. PAPER_364 — ALICE multiplicity centrality (original CP4 #8)
 3. PAPER_969 — Expanded 26D Ramanujan $S_{26}^{(k)}$
 4. ALICE Collaboration — Pb-Pb at $\sqrt{s_{NN}} = 5.02$ TeV
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Cross-Links

Related Paper	Relationship
PAPER_364	Original ALICE multiplicity class
PAPER_969	$S_{26}^{(k)}$ expansion

Related Paper	Relationship
PAPER_970	QGP vacuum density
PAPER_975	Triadic validation of ALICE

Calibration Constants

Constant	Symbol	Value	Validation Domain
A	—	2.0	Normalization
α	—	1.2	Centrality exponent
$\sqrt{s}$ exponent	—	0.156	Energy scaling
$[SSq]$	—	0.57	String coupling

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
$dN/d\eta$ centrality shape	Power-law with $S_{26}^{(k)}$	Consistent
Energy scaling	$\sqrt{s}^{0.156}$	ALICE-matched
$S_{26}^{(k)}$ modulation	Ramanujan correction	Novel

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Heavy-Ion Collisions (ALICE Centrality)

§A.2 Core Equation

$$\frac{dN_{\text{ch}}}{d\eta} = A \cdot \sqrt{s}^{0.156} \cdot (1 - c/100)^\alpha \cdot S_{26}^{(k)}$$

§A.3 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ vacuum density $\rightarrow S_{26}^{(k)} \rightarrow$ QGP formation $\rightarrow$ hadron multiplicity $\rightarrow$ ALICE $dN/d\eta$

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$S_{26}^{(k)}$ normalizes the multiplicity — VDS is the QGP production amplitude.

§B.2 DVP

Centrality selects overlap geometry — DVP modes in the overlap zone set the multiplicity.

§B.3 BSH

Peripheral collisions access BSH surface modes; central collisions access bulk.

§B.4 Summary

Metric	Value	Status
ALICE energy	13.6 TeV (Run 3)	Current
Centrality range	0-80%	Full
$[SSq]$	0.57	Calibrated